

supervisor	
signature	

* You must write your answer in English. Otherwise, penalty may be applied.

1. (26 points) Fill in the blanks (a)~(m) with the most appropriate English word(s).

2. (9 points) List three main benefits of data encapsulation and explain each benefit in sufficient detail.

benefits	explain each benefit in sufficient detail.
(i)	[use less than 10 English words]
(ii)	[use less than 15 English words]
(iii)	[use less than 15 English words]

([Use less than 15 English words.])

([Use less than 15 English words.])

(ii) In C/C++, dangling pointers can be a problem and dangerous. Why? Explain.
([Use less than 15 English words.])

([use less than 15 English words] _____)

(ii) What is the main purpose of using the `assert()` function?
([use less than 7 English words])

(iii) Fill in the empty blank with appropriate English words in the following sentence. Your answer should contain sufficient detail.

In order to achieve the purpose, **e** in the **assert(e)** function call should be a condition that ([use less than 10 English words])

6. (6 points) In project 1, we designed an IIKH (Interactive Intelligent Kitchen Helper) program. IIKH is supposed to contain **Recipe** class and **RecipeDB** (i.e. Recipe database) class. What is the main difference between **Recipe** class and **RecipeDB** class? Explain.

(i) **Recipe** class is supposed to () ← use less than 10 English words

(ii) **RecipeDB** class is supposed to () ← use less than 10 English words

7. (6 points) ADT (Abstract Data Type) is called abstract. Why? Answer by filling in the empty blank of the following sentence.

ADT is called abstract because ([use less than 12 English words]).

8. (15 points) Write a C++ program that performs file-copying from source binary file "**src.bin**" to destination file "**des.bin**".

Do not use a function (or a command) that performs file-copying operation directly. Instead, you are supposed to read the entire data from the file "**src.bin**" and write the data to the file "**des.bin**" for file-copying. Insert your code in the following empty box.

```
#include <iostream>
#include <fstream>
using namespace std;
int main() {
```

.....

```
}
```

9. (18 points) (1) Following C++ code has five errors. Which lines are wrong? How can you modify the code to correct them?

Assume that line 16 and line 17 are correct. 'line#' means line number.

(i) line#: (), corrected code: () (ii) line#: (), corrected code: ()

(iii) line#: (), corrected code: (), (iv) line#: (), corrected code: ()

(v) line#: (), corrected code: ()

(2) What will be the screen output (i.e. the result generated by cout) of the program after you make the correction in (1)?

Write your answer in the [screen output] empty box below.

```
1 : #include <iostream>
2 : using namespace std;
3 : class Rect {
4 : public:
5 :     int w,h;
6 :     Rect(int x, int y): w(x), h(y) { cout << "c1\n"; }
7 :     Rect(const Rect& x) { w=x.w; h=x.h; cout << "c2\n"; }
8 :     int area() { return w*h; }
9 : };
10:
11: void func(Rect obj, int y) {
12:     cout << "f3\n";
13:     obj.h = y;
14: }
```

```
15: int main() {
16:     Rect a(5,6);
17:     Rect *b, *c;
18:     b = a;
19:     c = new Rect(7,8);
20:     func(a,9);
21:     Rect d(a);
22:     b.w = 9;
23:     cout << "=====" << endl;
24:     cout << a.area() << endl;
25:     cout << b.area() << endl;
26:     cout << c.area() << endl;
27:     cout << d.area() << endl;
28:     delete b;
29:     delete c;
30:     return 0;
31: }
```

[screen output]
Write your answer
for (2) here.